

Case study on the TSI flagship projects: EU Supervisory Digital Finance Academy (2022-2023)

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Directorate-General for Structural Reform Support

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1. BACKGROUND INFORMATION

This case study presents a flagship project in the area of digital finance: the EU Supervisory Digital Finance Academy (EU-SDFA), a newly established academy that offers online and in-person courses for employees of national supervisory authorities. It is a multi-country project with 32 multi-country child projects, i.e. 32 central banks or financial supervisory authorities from 25 Member States who send employees to attend the EU-SDFA's modules. The technical details of these projects are provided in the table below.

The findings presented in this case study are based on desk research, analysis of monitoring data from DG REFORM, as well as 15 interviews with representatives of the European Commission, beneficiary authorities, national coordinating authorities and technical support providers.

TABLE 1. CHARACTERISTICS OF THE PROJECTS SELECTED FOR THE CASE STUDY

PROJECT ID	PROJECT TITLE	POLICY AREA	COUNTRY	TOPIC	BENEFICIARY AUTHORITY	FUNDING SOURCE	STATUS	TYPE	DELIVERY METHOD
TSI2022/ MCP/140	Flagship: EU Supervisory Digital Finance Academy: Strengthening Supervisory Capacity in Innovative Digital Finance	B5 - Financial Sector & Access to Finance	25 countries	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Multiple	TSI 2022, TSI 2023	On the ground	Multi-country parent	Multi-country component
TSI2022/ MCC/267	EUSDFA - AT - FMA	B5 - Financial Sector & Access to Finance	Austria	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Austrian Financial Market Authority (FMA)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/268	EUSDFA - CY - CBC	B5 - Financial Sector & Access to Finance	Cyprus	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Central Bank of Cyprus	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/269	EUSDFA - CY - CYSEC	B5 - Financial Sector & Access to Finance	Cyprus	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Cyprus Securities & Exchange Commission	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/270	EUSDFA - CZ - CNB	B5 - Financial Sector & Access to Finance	Czech Republic	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Czech National Bank	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/271	EUSDFA - DE - Bafin	B5 - Financial Sector & Access to Finance	Germany	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Bundesanstalt für Finanzdienstleistungsaufsicht (Federal Financial Supervisory Authority, BaFin)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/272	EUSDFA - EE -FI	B5 - Financial Sector & Access to Finance	Estonia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Estonian Financial Supervision and Resolution Authority (Finantsinspektsioon)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/273	EUSDFA - EL - Bank of Greece	B5 - Financial Sector & Access to Finance	Greece	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Bank of Greece	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/274	EUSDFA - EL - HCMC	B5 - Financial Sector & Access to Finance	Greece	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Hellenic Capital Market Commission (HCMC)	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/275	EUSDFA - ES - CNMV/BDE/DG SEGUROS	B5 - Financial Sector & Access to Finance	Spain	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Comision Nacional del Mercado de Valores Banco de España DG Seguros	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/276	EUSDFA - FI - FINFSA	B5 - Financial Sector & Access to Finance	Finland	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Finnish Financial Supervisory Authority (FIN-FSA)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/277	EUSDFA - HR - Croatian National Bank	B5 - Financial Sector & Access to Finance	Croatia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Croatian National Bank (CNB)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/278	EUSDFA - HR - Hanfa	B5 - Financial Sector & Access to Finance	Croatia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Croatian Financial Services Supervisory Agency (Hanfa)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/279	EUSDFA - IE - Central Bank of Ireland	B5 - Financial Sector & Access to Finance	Ireland	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Central Bank of Ireland	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/280	EUSDFA - IT - Banca d'Italia	B5 - Financial Sector & Access to Finance	Italy	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Banca d'Italia	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/281	EUSDFA - IT - CONSOB	B5 - Financial Sector & Access to Finance	Italy	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	CONSOB	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/282	EUSDFA - LU - CSSF	B5 - Financial Sector & Access to Finance	Luxembourg	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	CSSF (Commission de Surveillance du Secteur Financier)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/283	EUSDFA - MT - MFSA	B5 - Financial Sector & Access to Finance	Malta	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Malta Financial Services Authority	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/284	EUSDFA - NL - AFM	B5 - Financial Sector & Access to Finance	Netherlands	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Autoriteit Financiële Markten (AFM)	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/285	EUSDFA - PL - UKNF	B5 - Financial Sector & Access to Finance	Poland	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	The Polish Financial Supervision Authority – UKNF	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/286	EUSDFA - PT - CMVM	B5 - Financial Sector & Access to Finance	Portugal	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Comissão do Mercado de Valores Mobiliários (CMVM)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/287	EUSDFA - RO - ASF	B5 - Financial Sector & Access to Finance	Romania	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Financial Supervisory Authority	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/288	EUSDFA - SE - FI (Swedish FSA)	B5 - Financial Sector & Access to Finance	Sweden	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Finans-inspektionen	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/289	EUSDFA - SK - NBS	B5 - Financial Sector & Access to Finance	Slovakia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	National Bank of Slovakia	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/290	Supervisory Digital Finance Academy: Bank of Slovenia	B5 - Financial Sector & Access to Finance	Slovenia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Bank of Slovenia	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/291	EUSDFA - DK - DFSA	B5 - Financial Sector & Access to Finance	Denmark	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Danish Financial Supervisory Authority	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/292	EUSDFA - FR - ACPR	B5 - Financial Sector & Access to Finance	France	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Autorité de contrôle prudentiel et de résolution (ACPR)	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/293	EUSDFA - LT - BL	B5 - Financial Sector & Access to Finance	Lithuania	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Bank of Lithuania	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/294	EUSDFA - LV - FCMC	B5 - Financial Sector & Access to Finance	Latvia	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Financial and capital market commission	TSI 2022	On the ground	Multi-country child	Multi-country component

TSI2022/ MCC/295	EUSDFA - NL - DNB	B5 - Financial Sector & Access to Finance	Netherlands	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	De Nederlandsche Bank (DNB) Dutch Central Bank	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/296	EUSDFA - PT - BP	B5 - Financial Sector & Access to Finance	Portugal	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Banco de Portugal	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2022/ MCC/297	EUSDFA - RO - NBR	B5 - Financial Sector & Access to Finance	Romania	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	National Bank of Romania	TSI 2022	On the ground	Multi-country child	Multi-country component
TSI2023/ MCC/356	EUSDFA - BG - FSC	B5 - Financial Sector & Access to Finance	Bulgaria	Bank supervision and resolution, Capital Markets Union, Digital/Information Communications Technology (ICT), Insurance and pensions	Financial Supervision Commission	TSI 2023	On the ground	Multi-country child	Multi-country component

2. REFORM CONTEXT

2.1. Key policy problems and support needed

The EU Supervisory Digital Finance Academy (EU-SDFA), a TSI flagship initiative, was set up to help financial supervisory authorities cope with the risks and opportunities associated with the use of **advanced technologies in the financial sector**.

Digital finance, or fintech, is the term used to describe the impact of new technologies on the financial services industry.[1] These new technologies present great **opportunities to deliver new and traditional financial products and services**. As laid out in the SDFA's own e-book on 'Digital Finance in the EU', artificial intelligence (AI), machine learning (ML), blockchain or distributed ledger technologies, big data analytics, and cloud computing have brought about an unprecedented level of innovation in financial services.[2] Financial institutions, consumers, companies, and supervisory authorities alike benefit from easier and more inclusive **access** to financial services, a more **competitive** landscape, and a greater **choice** of financial products. The rise of online shopping, mobile banking, contactless payments, and cryptocurrencies illustrate the innovation these new technologies have made possible and their enthusiastic take-up by the general public. The COVID-19 pandemic has spurred the trend, increasing the use of financial applications in Europe by 72% within a week.[3] New technologies have also allowed financial institutions to increase the efficiency of their internal services.

Yet at the same time, the use of advanced technologies in the financial services industry presents **risks**. At an individual level, the use of AI-based applications can lead to biases, discrimination, and financial exclusion. At an organisational level, institutions are heavily relying on third party systems, and are increasingly vulnerable to ICT risks. These include cybersecurity threats (e.g. malware, viruses, hacking, phishing attacks, ransomware), data loss (unauthorised access to sensitive data), system downtime (outages disrupting business operations), software vulnerabilities (allowing hackers to gain access, manipulate data, or disrupt systems), human error (e.g. accidental edits), or insider threats (e.g. employees stealing data).[4] At a global level, digital finance has exposed the vulnerability of the whole financial sector.

The speed with which these new technologies and the Fintech ventures using them develop, and the magnitude of the risks and challenges associated with them raise a **need for regulating the new tools and technologies finance.** At the EU-level, the Commission is taking steps to make this Europe's "Digital Decade", aimed at strengthening the EU's digital sovereignty and setting standards instead of following those of others.[5] In September 2020, the European Commission adopted a digital finance package, which included a digital finance strategy as well as two specific legislative proposals on crypto-assets and digital resilience. This package is introduced in section 2.2.[6] Most beneficiary authorities interviewed for this study reported

[1] European Commission n.d. "Overview of digital finance".

[2] Guedel, Alice and Sciascia, Giuseppe. "The State of Digital Finance in Europe". In: Beck, Thorsten; Giani, Leonard and Sciascia, Giuseppe (Eds): "Digital Finance in the EU: drivers, risks, opportunities. The EU Supervisory Digital Finance Academy's First Year e-Book", p.14-15.

[3] European Commission 2020. "A Digital Finance Strategy for Europe", p.2.

[4] Chopskie, Edward. 20 July 2023. "What is Dora and why is it critical."

[5] European Commission. "A Europe fit for the digital age." Accessed 8 June 2024.

[6] European Commission. Communication from the Commission to the European Parliament, the council, the European Economic and Social Committee and the Committee of the Regions on a Digital Finance Strategy for the EU. Brussels, 24 Sep 2020, COM(2020) 591 final.

needing help – and receiving it through the Supervisory Digital Finance Academy – to implement the new legislation, in particular, the "Regulation on the Markets in Crypto-Assets" (MiCA).

In addition to implementing these EU-wide strategies, some Member States are advancing their own strategies in the area of digitalisation and the risks and opportunities arising from new technologies. For instance, the Belgian region of Wallonia[7], Croatia[8], Denmark[9], Estonia[10], Germany[11], the Netherlands[12], Portugal[13], Spain[14] have advanced national digitalisation strategies, covering the impact of digitalisation on various areas. Others have focused on digital skills: Hungary[15], Italy[16], Ireland[17], Latvia[18], Malta[19], Portugal, Slovakia[20], and Sweden have devised digital skills strategies (including digital skills in education, industry, or employment). Still others are devising specific artificial intelligence (AI) and cybersecurity strategies. Denmark,[21] Latvia,[22] and Poland,[23] and Slovakia[24] have devised national cybersecurity strategies while Austria[25], Bulgaria[26], Denmark[27], Finland,[28] Italy[29], Lithuania[30], Malta[31], Portugal[32] and Sweden[33] have put in place an AI strategy.

2.2. Adoption and implementation of administrative, institutional and structural reforms

As noted above, the European Commission adopted a digital finance package in September 2020, which included a **digital finance strategy.** The strategy set out four main priorities: (i)

[7] Digitalwallonia.be. "Digital Wallonia 2019-2024." Accessed 8 June 2024

[8] Rdd.gov.hr. "Digital Croatia Strategy for the period until 2032." Accessed 8 June 2024.

[9] Ministry of Foreign Affairs of Denmark. 19 Feb 2018. "The Danish Government Presents the Digital Growth Strategy". Accessed 8 June 2024

[10] Republic of Estonia. Ministry of Economic Affairs and Communications. "Digital Agenda 2030." Accessed 8 June 2024

[11] Federal Ministry for Economic Affairs and Climate Action. 1 April 2016. "Digital Strategy 2025". Accessed 8 June 2024

[12] Nederland Digitaal. 22 June 2021. "The Dutch Digitalisation Strategy 2021"

[13] Portugaldigital.gov.pt. 5 March 2020. "Portugal Digital".

[14] Espana digital. "Digital Spain Development." Accessed 8 June 2024.

[15] Digitalsjoletprogram.hu. 30 June 2016. "Digital Education Strategy of Hungary." Accessed 8 June 2024.

[16] Republicadigitale.gov.it. "Republica digitale". Accessed 8 June 2024.

[17] Government of Ireland. "Technology Skills 2022. Ireland's Third ICT Skills Action Plan". Accessed 8 June 2024.

[18] Digital Skills & Jobs Platform. National Strategies. "Latvia - Digital transformation Guidelines 2021-2027." Accessed 8 June 2024.

[19] Gvern Ta'Malta. "National eSkills Strategy 2022 - 2025." Accessed 8 June 2024.

[20] Mirri.gov.sk. "2030 Digital Transformation Strategy for Slovakia". Strategy for transformation of Slovakia into a successful digital country

[21] The Danish Government. "The Danish National Strategy for Cyber and Information Security." Accessed 8 June 2024

[22] Ministry of Defence Republic of Latvia. "Latvia approves new Cyber Security Strategy 2019-2022." Accessed 8 June 2024.

[23] Ministry of Digital Affairs. "National Framework of Cybersecurity Policy of the Republic of Poland for 2017-2022."

[24] National Security Authority. "National Cyber Security Strategy"

[25] Bundesministerium Klimaschutz, Umwelt, Energie, Mobilität, Innovation und Technologie. "Artificial Intelligence Mission Austria 2030." Accessed 8 June 2024.

[26] MTC Government Bulgaria. Oct 2020. "Concept for the Development of Artificial Intelligence in Bulgaria until 2030." Accessed 8 June 2024.

[27] Agency for Digital Government. "The Danish National Strategy for Artificial Intelligence." Accessed 8 June 2024;

[28] Ministry of Economic Affairs and Employment of Finland. "Artificial Intelligence 4.0 programme accelerates business digitalisation."

[29] Docs.italia.it. "Strategic Programme on Artificial Intelligence 2022 - 2024". Accessed 8 June 2024.

[30] Eimin.lrv.lt. "Lithuanian Artificial Intelligence Strategy." Accessed 8 June 2024.

[31] Malta.AI. Oct 2019. "Malta. The Ultimate Launchpad. A Strategy and Vision for AI in Malta 2030."

[32] InCode2030. "AI Portugal 2030. Portuguese National Initiative on Digital Skills". Accessed 8 June 2024.

[33] AI Sweden. "An AI Strategy for Sweden". Accessed 8 June 2024

removing fragmentation in the Digital Single Market (including through the use of digital identities), (ii) adapting the EU regulatory framework to facilitate digital innovation (iii) promoting a data-driven finance and addressing the challenges and risks with digital transformation and (iv) enhancing the digital operational resilience of the financial system.

In addition to the digital finance strategy, the package included two specific legislative proposals: one on crypto-assets and one on digital resilience. The **"Regulation on the Markets in Crypto-Assets" (MiCA)** applies to firms that are active in the crypto-asset market.[34] MiCA entered into force in June 2023. Its first parts (provisions regarding asset-referenced tokens and e-money tokens) will become applicable from 30 June 2024. MiCA regulates crypto-assets and related services and activities that were not regulated by existing financial services regulation. It aims to simplify the post-trading of crypto-assets within the EU and make them more secure.

MiCA addresses four main risks that have appeared in the crypto-asset market. First, MiCA holds that prospective customers and crypto-asset holders need to be informed about the risks of the products they hold or intend to purchase. Second, to reduce the risk of fraud, MiCA introduces operational requirements for those providing crypto-asset services (e.g. trading venues or wallets). Third, to avoid hacks and bugs in the blockchain, MiCA requires the establishment of adequate IT security procedures and systems to guard against cyber risks and IT failures. Fourth, to mitigate risks of money laundering, terrorist financing and sanctions circumvention, MiCA includes crypto-asset service providers in the list of 'obliged entities' under the anti-money laundering framework (AML). This means that they have to comply with EU laws to combat money laundering and the financing of terrorism (AML/CFT).[35]

The second draft legislative proposal which had been included in the digital finance strategy has a wider scope: the **"Digital Operational Resilience Act" (DORA)** applies to almost the entire financial industry in the EU. It was adopted by the Parliament and the Council in December 2022.[36] **DORA aims to prevent and mitigate cyber threats, and to prepare companies and organisations for large-scale attacks on the European financial sector.** It addresses companies and organisations that operate in the financial sector, such as banks, insurance companies and investment firms as well as critical third parties that provide services to them, such as cloud platforms or data analytics services. DORA sets uniform requirements for the security of network and information systems of these players to make sure they can withstand, respond to ICT-related disruptions and threats.[37]

At present, supervisory authorities across the 27 Member States are taking steps to implement the new legislation. Yet in order to do so, supervisors need an in-depth understanding of not only the legislation itself but also of the technologies underlying the phenomena it regulates, and the associated risks. As one beneficiary authorities explained in the case of MiCA, "it is one thing to understand the regulation, but it is another thing to understand what it is talking about" – in this case, crypto-assets. This is the need that is addressed by the Supervisory Digital Finance Academy.

[34] Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets, and amending Regulations (EU) No 1093/2010 and (EU) No 1095/2010 and Directives 2013/36/EU and (EU) 2019/1937
[35] European Commission. Crypto-assets. „What the EU is doing and why".
[36] Regulation (EU) 2022/2554 of the European Parliament and of the Council of 14 December 2022 on digital operational resilience for the financial sector and amending Regulations (EC) No 1060/2009, (EU) No 648/2012, (EU) No 600/2014, (EU) No 909/2014 and (EU
[37] Council of the European Union. 28 Nov 2022. Press release "Digital finance: Council adopts Digital Operational Resilience Act"

3. DESIGN AND EXECUTION OF TSI PROJECTS

The EU-SDFA is a year-round Academy that offers a range of modules designed to help financial supervisory authorities deal with the risks and opportunities associated with the use of advanced technologies in the financial sector. It was established by the European Commission (DG REFORM), in collaboration with the **Florence School of Banking and Finance** (FBF) within the European University Institute (EUI), and the three European Supervisory Authorities (ESAs), that is, the European Banking Authority (EBA), the European Securities and Markets Authority (ESMA) and the European Insurance and Occupational Pensions Authority (EIOPA). These four institutions – the FBF, EBA, ESMA, and EIOPA – are the technical support providers responsible for the overall project implementation under the steering and coordination of the Commission DG REFORM. The EU-SDFA was established for a period of **three years**, starting in the academic year of 2022/23. The second academic year, 2023/24 is nearly completed; the third and final year will begin in the autumn of 2024. The beneficiary authorities are the national financial supervisory authorities who send staff to attend the various training modules and activities offered within the project.

The expected outcome of the EU-SDFA is "to enhance a **robust supervisory framework** on an EU-wide scale, contributing to the dissemination of advanced, future-proof practices among national authorities, as well as to the exchange of expertise both on a cross-sectoral and cross-border basis."

The EU-SDFA offers a comprehensive training curriculum including in-person training courses at different levels, foundational, and advanced modules, but also on-line training courses and workshops. While most are taught by the FBF and the ESAs, the EU-SDFA also brings in instructors from the industry, other universities, the European Commission, global regulatory institutions, and national financial supervisory authorities themselves. The latter are brought in to encourage peer-learning: employees of national financial supervisory authorities who have attended courses in the past are sometimes asked to return as instructors and share their experiences or best practices in various areas of digital finance.

There are three types of training activities, which are held throughout the year: residential trainings, online modules, and workshops. **The residential training** brings financial supervisors together in person. These courses are held in Florence, at the EUI's premises, normally for a duration of three and a half days. Each day, participants spend 6 to 7 hours in class, for a total of around 24 hours of training per week. In addition to the classes, the FBF organises dinners, keynote speeches and social activities allowing participants to build an informal network with fellow supervisors. Residential training courses are generally limited to 35 participants in total, and these typically include officials from each supervisory authority participating to the EU-SDFA. The residential training includes foundational blocks, as well as advanced blocks.

The **foundational blocks** (3.5 days) allow participants to acquire a fundamental knowledge of the main areas of digital finance, as well as the practical skills to enable them to effectively supervise the application of digital technologies in finance by credit and financial institutions, and to effectively use supervisory technology (SupTech). The structure rests on three building blocks. In the first block, participants explore the main technologies in digital finance, including AI, big data, and distributed ledger technology, the risks and opportunities these new technologies present, and their application in RegTech and SupTech. In the second block, participants learn about new EU regulation, including the European Digital Finance Strategy, MiCA, DORA, and the guidelines published by the ESAs. This block includes a session on NCA's best practices in the domain of digital finance, where participants from within the EU, and outside the EU share their own practical experiences supervising emerging technologies. The third block is a practical block, aimed to provide supervisors with the necessary skillset to operate in the

new supervisory framework in digital finance. In this block, participants practice their newly acquired knowledge of the new technologies and legislation in case studies and other practical exercises. [38]

The **advanced blocks** (3.5 days) delve deeper into some of the topics covered in the foundational course. In contrast to the foundational blocks, which are only delivered in person, some advanced blocks are delivered in person, and some are delivered online. For instance, in 2023/24 the EU-SDFA offered a range of advanced residential courses, each focussing on a different topic such as 'Cyber Risk', 'SupTech', 'Data-driven Business Models & Data Sharing in Finance', and 'Distributed Ledger Technologies / MiCAR'. In addition, in 2023/04 it offered online courses on 'MiCAR', 'DORA', and 'AI in the Financial Sector'. [39]

The **online modules** (9-10 hours over the course of three weeks) run over two weeks and focus on regulatory aspects. These modules allow participants to watch the training videos online, in their own time, and discuss course content during short live classes. For instance, participants might watch a video lecture on a topic such as MiCA, explaining the regulatory framework of the legislation, and then use the live classes to discuss any questions that arose from the lecture and to work on specific use cases along with colleagues from other countries. [40]

The **workshops** are the shortest modules, lasting a day and a half. These are also held in person – either in Brussels, Paris (EBA/ESMA), Frankfurt (EIOPA), or Florence (EUI) – and a limited number of seats is also offered for online attendance. In 2023/04, the EU-SDFA offered advanced workshops on 'Executive Seminar Digital Finance Supervisory Implications', 'Emerging Digital Finance Issues Workshop (EU-SDFA Lab in Digital Finance): Harnessing the power of the cloud for the supervisory good', and 'Policy Research Workshop on Digital Finance'. [41] In addition, ESMA convened a course on 'Digitalisation and consumer protection' in Paris; EIOPA convened an 'AI Supervision workshop' in Frankfurt, and the EBA convened a 'MiCAR Supervision' workshop in Paris. [42]

3.1. Relevance and coherence

All beneficiary authorities interviewed for the purposes of this study saw the courses offered by the EU-SDFA as very helpful and well-tailored to their needs. Almost all interview participants from beneficiary authorities consulted for this study attended EU-SDFA modules themselves. Some of them pointed out that there are many courses on new technologies, such as artificial intelligence or machine learning on the web. Yet none of these publicly available courses address the very specific needs of a supervisory authority. There is no other intervention, at the EU level, or at the national level for capacity-building training courses for financial supervisors. The EU-SDFA is unique in that it allows financial supervisors to acquire not just a general knowledge of these new innovative technologies but a knowledge that is **targeted to their role as financial supervisors**. The main motivation that prompted national financial supervisory authorities to send staff to the workshops was to learn skills that are, or will be needed in their job, to give

[38] EU-SDFA. "Foundational blocks. Programme." Accessed 23 July 2024.

[39] EU Supervisory Digital Finance Academy. Advanced courses for the Academic Cycle 2023-2024. "Advanced Online Modules", p.2-8. Accessed 23 July 2024.

[40] EU Supervisory Digital Finance Academy. Advanced courses for the Academic Cycle 2023-2024. "Residential Training Weeks", p.9-12. Accessed 23 July 2024.

[41] EU Supervisory Digital Finance Academy. Advanced courses for the Academic Cycle 2023-2024. "Workshops", p.13-16. Accessed 23 July 2024.

[42] EU Supervisory Digital Finance Academy. "Workshops". Accessed 23 July 2024.

them background knowledge that will be helpful in their role and to help them understand new regulation before it comes into effect.

Understanding and implementing new legislation in digital finance

Beneficiary authorities reported that the EU-SDFA provided their staff with the background knowledge they needed to implement MiCA and DORA. In particular, financial supervisors reported that the EU-SDFA filled gaps in institutional knowledge about blockchain, decentralised finance (DeFi), and crypto-assets – knowledge that is crucial to understand both the need for the EU to regulate the crypto-market and regulation itself. Financial supervisors also reported that the case studies and example cases were very helpful to understand the scope of the legislation and to prepare them to implement it in practice.

This was also the area in which national financial supervisory authorities saw the greatest potential for improvement. Even though the EU-SDFA prioritised training for MiCA and DORA, allowing more spaces for both in-person and online training on these two pieces of legislation the **demand for training on MiCA and DORA still surpassed the allotted spaces**. In particular, beneficiary authorities would have appreciated more spaces in MiCA training courses to be able to train more of their staff. Many of those who did attend a MiCA training course also said that they would have preferred to take even more time or go into even more depth in those sessions. All reported that the courses were very helpful but the size of the gaps in their understanding of the legislation and its application were so large that a few more hours and more practical examples would have helped. Future iterations of the EU-SDFA may consider expanding sessions on new legislation.

More generally, beneficiary authorities reported that the EU-SDFA helped financial supervisors in their countries gain an in-depth understanding of new technologies, including those around artificial intelligence, preparing individuals and institutions for future applications. This contributes to the EU's objective of a Europe fit for the digital age.

3.2. Efficiency

A cost-efficient project

With an overall available call budget of EUR 5 510 000[43], the cost per participant has been relatively low in comparison to similar forms of provision. In its first two years, the courses organised by the EUI have reached at least 1632 participants (see section 3.3). Assuming that the number of participants who will be reached in year 3 is similar to that in year 2, i.e. somewhere between 944 and 949 individuals, (429 participants completing the residential trainings; 440 completing online modules, and 75-80 completing the EUI workshops)[44], it can be expected that the EUI-based courses will have reached at least 2576 individuals (1632 in years 1 and 2 + 944 in year 3). Given that the number of participants has risen between year 1 and 2, and more importantly, given that these numbers do not include the courses organised by the ESAs, for which we lack statistics, this is a very conservative estimate.

Using this very conservative estimate, then, the cost per participant is at or (very likely) below EUR 2 138 (EUR 5 510 000 / 2576 participants).[45] While the unique focus of the EU-SDFA makes

[43] European Commission. Technical Support Instrument (TSI). Invitation to submit a proposal. EU Supervisory Digital Finance Academy (TSI-2022.EUSDFA-IBA). 17 Feb 2022, p.7.
[44] CSES Consultations with EUI, 10 June 2024.
[45] The unique nature of the EU-SDFA with a curriculum that is tailor-made for financial supervisors in the EU makes it difficult to compare it with other programmes. However, one-week courses in the broad field of digital finance, AI or financial regulation tend to be much more expensive. For example, the University of Oxford offers a five-day "Bank

it difficult to compare the price with other summer schools, this is a very low fee for an in-person summer school in a top university.

Management

Nearly all course participants interviewed for this study reported that the overall **quality** of the courses was excellent, and that the EU-SDFA was **very well managed**.

Almost all beneficiary authorities consulted for this study reported that **communication** with DG REFORM as well as the technical support providers, in particular, the EUI, was excellent. All interviewees reported that the courses, and particularly the residential courses at the EUI, were very well managed. The team managing the course on-site were exceptionally responsive and helpful with any queries. Upon arrival in Florence, course participants reported, everything was sorted; the accommodation, the modules, and the extracurricular activities were so well organised that they were able to focus entirely on the course content.[46]

Drivers of a successful implementation

Course participants reported a few aspects that contributed to the efficient implementation of the project. The main aspect almost all course participants noted was the opportunity to get to know fellow financial supervisors in an **informal** setting. Dinners and extracurricular activities contributed to a relaxed atmosphere allowing participants to build **networks** with financial supervisory authorities across Europe. A few participants reported staying in touch with financial supervisors from other EU countries whom they met at the EU-SDFA in Florence after the end of the course.

Beneficiary authorities also appreciated the variety of **teaching formats**. The EU-SDFA allows participants to take courses online, as well as in person. The majority of beneficiary authorities reported that the in-person courses, in particular, the longer, 3.5 day in-person courses in Florence were more efficient than the online courses as they allowed participants to take time off work and focus entirely on the course. Participants also reported that the opportunity to connect with other financial supervisors and to ask questions enhanced their learning experience.

However, beneficiary authorities were grateful to also have the option of sending staff to attend **online** training sessions. One participant reported that, as a parent and a carer, they were unable to travel abroad, which meant that they were not able to attend the residential training course in Florence. However, because some training sessions were offered online, they were still able to participate. As such, the online modules add to the inclusive nature of the EU-SDFA. However, the participant saw room for improvement: At present, online modules do cover the issues beneficiary authorities saw as most urgent – MiCAR, and DORA – but they do not cover

Governance Programme". In 2025, the tuition alone is GBP 8,500 (around EUR 10,210), plus accommodation (GBP 1,500 or around EUR 1,80). Online courses at the University of Oxford are somewhat cheaper: A three-week 'Artificial Intelligence Programme' is GBP 2,300 or around EUR 2,760). Similarly, the Frankfurt School of Finance and Management offers a five-day summer academy with different tracks including 'Inclusive & Digital Finance' and 'Risk Management & Analytics for Inclusive Financial Services.' Here, the tuition alone is EUR 5,700. Another example is the Barcelona School of Economics which offers a one-week summer school in 'Banking Regulation and the Future of Money' with a tuition fee (only) of EUR 1,150. Finally, the London School of Economics offers a range of summer school courses, including one on 'International Financial Regulation in the Digital Area' GBP 2,950 (or around EUR 3,550) not including accommodation). In addition to being generally more expensive these summer schools will only benefit direct participants. They do not make their course material available to the general public as does the EU-SDFA. Sources: Said Business School, University of Oxford. "Oxford Bank Governance Programme." Accessed 22 November 2024; Frankfurt School of Management and Finance. "Frankfurt School Summer Academy." Accessed 22 November 2024; Barcelona School of Economics. "Banking Summer School." Accessed 22 November 2024; The London School of Economics and Political Science. "Summer School 2024 Fees." Accessed 22 November 2024.

[46] CSES Consultations.

the full range of topics taught in in person. A few beneficiary authorities reported that it would be helpful to have an online option for the whole foundational training.

Another aspect that was reported to be efficient was the division of course materials into elementary and advanced modules, and the new guidance on **prerequisites** for any specific course. In the first year of the EU-SDFA, a coordinating authority as well as a technical support provider consulted for this study reported some hiccups in selecting the right staff to attend the various courses. Sometimes, the lack of specific information on the courses led to national coordinating authorities sending financial supervisors to courses that were too easy or (more often) too difficult for them. Responding to this feedback, the technical support providers now publish specific requirements for each module in the courses' booklets, during dedicated information sessions open to all participating authorities and – upon request – via email. This has reduced mismatches. Nonetheless, one coordinating authority reported that, even though the information was very helpful, they still sometimes found it difficult to choose between staff who would like to attend and would therefore benefit from an even more detailed information about the content of the courses, the prerequisites, and the type of participant each course is aimed at.

Barriers and room for improvement

One area where beneficiary authorities saw room for improvement was the **intensity** of the in-person courses and the **flow of information** in the run-up to the in-person seminars. Regarding the intensity of the foundational course in particular, some participants reported they would have preferred stretching the material over a week rather than 3.5 days in order to have more time for each session. Regarding the flow of information, the main concern was the timing: Three beneficiary authorities reported that the detailed information for the in-person courses in Florence, in particular, information regarding the preferred arrival times and travel instructions reached them relatively late. Receiving this detailed information earlier on – ideally, around two months before the respective course – would make it easier for participants to book their flights earlier on, when they are still relatively cheap. Responding to the expressed need to receive more information earlier on, the EU-SDFA has already made some changes: it now offers information sessions well in advance of the courses. For instance, for the training sessions taking place between January and June 2024, the EU-SDFA organised an information session on 13 December 2023.

One beneficiary authority noted that they would also appreciate more assistance with travel expenses, since the amount provided by the EU funding was not always sufficient to cover the travel expenses to Florence. In addition, this same beneficiary authority noted that travel expenses for the shorter workshops were not covered. This restriction has meant that some departments in this beneficiary authority have not been able to send staff to Florence.

Two beneficiary authorities noted that they would have appreciated receiving more **course materials** ahead of the respective course. Given the small number of slots, participants generally saw it as a great privilege to be able to travel to Florence and to participate in the residential training. As such, many participants, in particular for whom the course material was challenging, reported that they would have liked to read it ahead of time. While it can be expected that all participants would benefit from self-study to prepare themselves for their courses, the opportunity to do so is particularly important for those who have less background knowledge – in other words, those who need the courses the most. Sharing course materials well in advance will also allow participants to make the most of the opportunities to connect with other course participants. The more prepared financial supervisors are the less time they will need for self-study during their stay in Florence, and the more time they will have for informal

conversations with other financial supervisors, building and strengthening these important informal ties and networks.

3.3. Effectiveness and sustainability

Two years into the programme, the EU-SDFA has reached all expected outcomes. The grant agreement specified four specific objectives, along with measurement units and targets.

Expected outcomes

The EU-SDFA's first specific objective was "**Consolidating expertise and technical knowledge in digital finance**, in particular the understanding of new innovative technologies and the risks emanating from the use of these technologies in the financial sector and the early identification of measures that could contribute to offset or minimise these risks."[47] The measurement unit suggested in the grant agreement was the number of participants trained per year, with a target of 240 trained participants/year in steady state (year 3). Two years into the programme, this target has been surpassed: As of June 2024, the SDFA has reached **at least 1632 supervisors** across the EU. In total, 724 individuals have completed the residential trainings (295 in year 1, and 429 in year 2), and 833 have completed online modules (393 in year 1, 440 in year 2). In addition, by the end of year 2, 75-80 individuals will have completed the EUI workshops.[48]

The second specific objective was "**Enhancing supervisory technologies and best practices,** with a specific aim at levelling the expertise relative to digital finance of different authorities within Member States."[49] The measurement unit suggested in the grant agreement was the number of knowledge exchange sessions per year, with a target of 6 knowledge exchange sessions in year one, and 9 knowledge exchange sessions in years 2 and 3. This target been surpassed in both year one and year 2: In year 1, 2022-23, the EU-SDFA offered not 6, but **15 training courses**, including foundational and advanced training weeks, as well as workshops (+9). In year 2 (2023-24), the EU-SDFA offered not 9 but **21 training courses**, including foundational and advanced training weeks, workshops, and online modules (+12). In total, as of June 2024, the EU-SDFA has conducted **36 training courses**.[50]

The third specific objective was "**Improving synergies between supervisory authorities, enabling peer learning, and increasing the level of knowledge** (but also to reduce supervisory costs and enhance harmonisation of supervisory practices)."[51] In 2023, the EU-SDFA was in line with the targets set in the grant agreement by reaching **32 authorities** (i.e. central banks and financial supervisory authorities) benefiting from and contributing to the project from **25 Member States**.

The fourth specific objective was "**Facilitating the transfer of expertise and knowledge** of the functioning and application of digital technologies in finance from expert industries to supervisory authorities".[52] The measurement unit suggested in the grant agreement was the number of knowledge outputs, with a target of 25 audio-visual outputs and 10-12 written outputs

[47] European Commission. "Technical Support Instrument (TSI) Application Form. Technical Description" (Part B), p.7. In: European Commission. 28 July 2022. "Grant Agreement 101092497 — EUSDFA", p.7
[48] CSES Consultations with EUI, 10 June 2024. The numbers for the EUI workshop are approximate because the last workshops for year 2 will only be held in September 2024.
[49] European Commission. "Technical Support Instrument (TSI) Application Form. Technical Description" (Part B), p.7. In: European Commission. 28 July 2022. "Grant Agreement 101092497 — EUSDFA", p.7
[50] EUSDFA 2024. Activities Calendar. Accessed 1 June 2024.
[51] European Commission. "Technical Support Instrument (TSI) Application Form. Technical Description" (Part B), p.7. In: European Commission. 28 July 2022. "Grant Agreement 101092497 — EUSDFA", p.8
[52] European Commission. "Technical Support Instrument (TSI) Application Form. Technical Description" (Part B), p.7. In: European Commission. 28 July 2022. "Grant Agreement 101092497 — EUSDFA", p.8

per academic year.[53] As of July 2024, the EU-SDFA's YouTube channel includes **30 videos**[54] (+5). In addition, the EU-SDFA has published an **e-book**, available on its website, that contains **12 chapters** (+2/4) on the main topics covered in the EU-SDFA's first year. Most chapters were written by the instructors based on teaching materials. The book is designed to complement the modules and to provide EU-SDFA participants with a reference document to consult after they have finished the respective course.

Use of course content in the supervisory authorities

According to feedback the EU-SDFA collected after its first year, between 9 and 10 out of 10 participants in each of the six weeks answered 'Yes' when asked whether they would be able to mobilise a substantial part of the knowledge they acquired in this training in their work. This reflects the data in our interviews, where all course participants who had participated in a course themselves reported using the skills and the information they learned in the EU-SDFA in their respective jobs. The nature of the technical support, the fact that it equips individuals with technical knowledge and practical ideas for how to solve problems makes it difficult to pinpoint the exact impact on the supervisory authorities' work. It is possible that the EU-SDFA's greatest achievements – such as, for instance, an institution taking steps to avoid large-scale cyberattacks or successfully countering one – will never reach public knowledge. However, all participants consulted for this study reported that they themselves and their colleagues who attended the seminars have gained a deeper understanding of the new technologies covered in the courses they attended, and that this knowledge had helped them carry out their work as regulators. For many, the most direct outcome was a deeper understanding of the new legislation in the field of digital finance. As such, the EU-SDFA has been instrumental in preparing supervisory authorities to implement MiCA and DORA.

Consequently, the **overall goal of enhancing a robust supervisory framework on an EU-wide scale has been met**. The EU-SDFA is successfully contributing to the dissemination of advanced, future-proof practices among national authorities, as well as to the exchange of expertise both on a cross-sectoral and cross-border basis.

Sustainability

The sustainability of the benefits of the project is supported by the provision of high-quality materials, which enables participants to reconsult them later and share them with colleagues. Most beneficiary authorities consulted for this study **disseminated the materials and/or the learning to other staff in their respective institutions**. Some course participants shared the course materials on Microsoft Teams channels or other platforms, others shared the main points they had taken away from their courses in team meetings or in informal conversations with colleagues. Still others prepared PowerPoint presentations focusing on the information that was most relevant to their respective institution, or unit. One beneficiary authority has put in place an 'implementation team' of EU-SDFA-trained financial supervisors tasked with putting new skills into practice in their financial supervisory authority.

While the lecture materials, such as PowerPoint slides and case study materials are only available to the respective participants, the EU-SDFA is also producing **publicly available content** for supervisors and other players in digital finance. This includes an e-book written by the EU-SDFA's instructors on the main topics they taught in the EU-SDFA's first year, as well as a YouTube channel.

[53] EUSDFA 2024. Activities Calendar. Accessed 1 June 2024.
[54] YouTube Channel. "EU Supervisory Digital Finance Academy." Last accessed 1 June 2024.

As of late July 2024, the EU-SDFA's **YouTube channel** comprises 30 training videos, mostly between 5 and 8 minutes long. Each video features a different instructor from the Commission, the academy, the European Supervisory Authorities, as well as the national supervisory authorities. Some videos introduce **innovative technologies** in digital finance, e.g. 'The use of technology for supervision' (Banco de España), 'Data science for authorities and central banks' (Dutch Central Bank), 'Natural language processing in the area of banking supervision' (European Central Bank), 'Machine learning for supervision' (Swedish Finansinspektionen), or 'Financial innovation and digital finance' (Swedish Finansinspektionen). Two videos focus on **supervisory technology** for regulators ('SupTech Solutions: Build v. buy' (University of Cambridge), 'SupTech strategy and innovation' (Bank of Latvia).[55]

Other videos discuss **EU legislation** in digital finance, e.g. 'Credit risk models & the AI Act' (University of Edinburgh Business School), "The implications of the AI Act for the financial sector' (European Commission). Still others present **risks**, e.g. 'Pitfalls in supervising digital finance' (European University Institute), or 'Digitalisation vulnerabilities' (European Securities and Markets Authority (ESMA), or **best practices**, e.g. 'Best practices for a Fintech programme' (Malta Financial Services Authority), or 'Spanish sandbox and risk management' (Banco de España). As of 1 June, views ranged between 77 for the most recent one and 290 for a video published 11 months ago.)

3.4. EU value added of the projects

The EU-SDFA exploits **synergies across Member States** by creating the opportunity for financial supervisors from different Member States to learn together and to learn from each other. Before the EU-SDFA was established the different supervisory authorities in the EU had fewer opportunities to interact, in particular, at lower levels, and across sectors. While the European Supervisory Authorities (EBA, EIOPA, and ESMA) have long facilitated interactions to ensure consistent financial supervision across the EU, these interactions tended to happen at higher levels of hierarchy and on a sectoral basis. The EU-SDFA has placed the network on a new level, reaching all levels of hierarchy, and all sectors. The EU-SDFA brings financial supervisors from different EU countries, different sectors, and different levels of hierarchy together in one room. This allows financial supervisors to exchange best practices, and to learn from one another. One beneficiary authority interviewed for this study reported that before the EU-SDFA was created, their institution rarely consulted with other financial supervisory authorities – even though, this individual believed, other institutions were probably faced with the same issues at the time. Any information exchange at the lower levels of hierarchy depended entirely on the initiative of individual supervisors and, if at all, was restricted to institutions in neighbouring countries were individual supervisors happened to know someone. The EU-SDFA has created a **knowledge hub** for national finance supervisors, institutionalising information exchange at a time where the risks and opportunities of digitalisation make collaboration indispensable. Given the small size of the financial supervisory community in the EU, contacts that have been established are likely to have a lasting impact.

Sharing experiences in a classroom setting

The EU-SDFA encourages financial supervisory authorities to learn from the experience of their counterparts in other EU countries by inviting national financial supervisory authorities who have got relevant experiences to share to attend modules as instructors. Course participants enjoyed hearing other supervisory authorities' experiences and **best practices in risk management,**

[55] YouTube Channel. "EU Supervisory Digital Finance Academy." Last accessed 23 July 2024.

such as the experience of a supervisory authority managing a large-scale cyberattack. These concrete examples illustrate how a risk can become reality and how it can affect entire institutions. As such, the EU-SDFA raises awareness of very real risks and encourages individuals and organisations to take steps to prepare for these risks. Moreover, the experience of financial supervisory authorities who have, in this case, dealt with a cyber-attack prepares national supervisory authorities in other countries for the eventuality of similar scenarios happening in their own institutions.

Course participants also benefitted from the opportunity to learn from supervisory authorities who are **pioneering new tools and technologies**. In particular, course participants valued the real-world examples from supervisors using new data sources and new techniques (or, 'SupTech') to improve their workflow or capture risks more precisely. For instance, participants enjoyed hearing how other supervisory authorities leverage the opportunities of artificial intelligence and big data analytics in central banking. Here, one example is a financial supervisory authority who, among other things, optimised data collection, moving away from expensive and error-prone processes in which individuals collected data manually in Excel files, and toward automated web scraping approaches.[56] Some supervisory authorities have recorded videos for the EU-SDFA's YouTube channel to share their experiences with SupTech strategy with an even wider public.

Sharing experiences in an informal setting

While these taught examples were seen as helpful and informative, the most frequently mentioned advantage of the EU-SDFA is that it gives financial supervisors the opportunity to **meet other financial supervisors in person and in an informal setting**. This informal setting creates EU value added by giving financial supervisors a space to share their own personal experiences with each other and to discuss common challenges. The design of the modules facilitates this network building by requiring course participants from different European countries to work together on **group projects** in which participants work together and see how other financial regulators approach specific tasks. In particular, participants enjoyed tasks in which they were presented with problems that other national supervisory authorities had faced in the past, hypothetical problems they could all face in the future, or test cases to apply a new legislation. The classroom setting allowed participants to explore the different options together, to consider advantages and disadvantages of various course of action, and to benefit from each other's ideas and experiences. These exercises were seen as a very effective way to prepare **individual financial supervisors to cope** with issues that are likely to come up in the future.

Adding to the sustainable nature of the EU-SDFA, a few participants of the in-person courses in Florence reported **using the network long after the actual course in Florence**. It can be assumed that this longer, in person course was particularly effective at allowing participants to build networks because it allowed participants to get to know each other over the course of a few days, both through group work and case studies, and through off-site social events.

Contribution to the Digital Transition

The EU-SDFA contributes to the **digital transition** in two main ways. First, it equips individual financial supervisors with the knowledge they need to effectively **implement EU legislation**, reducing security risks and helping meet the 2030 digital infrastructure targets. Second, it empowers individuals to **use the full potential of new technologies**, in particular, artificial

[56] Guedel, Alice and Sciascia, Giuseppe. "The State of Digital Finance in Europe". In: Beck, Thorsten; Giani, Leonard and Sciascia, Giuseppe (Eds): "Digital Finance in the EU: drivers, risks, opportunities. The EU Supervisory Digital Finance Academy's First Year e-Book", p. 68.

intelligence, by bringing them together with colleagues in other national authorities who have successfully integrated these new technologies in their workflows and who share their experiences in EU-SDFA modules.

CONCLUSION

As of early June 2024, the EU-SDFA has conducted 36 training courses for more than 1,600 supervisors from 37 financial supervisory authorities in 26 Member States. As such, **it has exceeded its targets.** This has been achieved at a very reasonable cost. According to our very conservative estimate, the cost of each course per participant is just above EUR 2 000, in addition to the free lectures available on the EU-SDFA's YouTube channel and, most importantly, the freely available handbook, which makes the content of the foundational block available to all, at no cost. The most direct impact reported by beneficiary authorities were the training courses on EU legislation regarding crypto-assets (MiCA) and ICT risks (DORA). The modules taught as part of the EU-SDFA, the exchange with other regulators, and the case studies they worked through in the classroom have **prepared supervisory authorities across the participating Member States to implement the new EU legislation.** As such, the EU-SDFA contributes to the successful implementation of EU regulation, which, in the case of MiCA, regulates crypto-assets to allow for innovation in a way that preserves financial stability and protects investors and, in the case of DORA, helps firms such as banks, stock exchanges, clearinghouses, and fintechs withstand ICT-related threats and disruptions, enhancing cybersecurity across the industry and guarding the EU against large-scale attacks.[57]

A multi-country flagship project

The nature of the call, a multi-country flagship project, was generally seen as an effective and efficient way to support financial supervisory authorities. The issue at hand – managing the risks and leveraging the opportunities of new technologies in digital finance – is a top EU priority. The EU-SDFA is closely tied to the EU Digital Strategy, in particular, to three of the four cardinal points of the 2030 Digital Compass. It contributes to 'A digitally skilled population and highly skilled digital professionals' by training individuals in one profession where digital skills are particularly important: Financial supervisors require an in-depth understanding of the new technologies to regulate those new technologies effectively. The EU-SDFA also contributes to the 'Digital transformation of businesses' by teaching financial supervisors to take full advantage of the benefits of modern technology to improve the efficiency and effectiveness of their own workflows. Finally, the EU-SDFA contributes to 'Secure and performant sustainable digital infrastructures' by enabling financial supervisors to implement much-needed EU-wide reforms that will, among other things, prepare the sector for attacks on the European banking sector.[58] As such, a flagship project is appropriate.

All beneficiary authorities consulted for this study reported a need to strengthen capacity in their supervisory authorities. As such, a multi-country project is appropriate. The fast pace of technological developments and the fact that some areas, such as crypto-assets, are now being regulated for the first time poses a challenge for many supervisory authorities. As one beneficiary authority interviewed for this study noted, they cannot always hire new people to keep up with the change. Therefore, training their own workforce is crucial for supervisory authorities to be able to effectively regulate and supervise the market in a high-risk, and high-opportunity digital age.

All beneficiary authorities consulted for this study saw clear value added in a multi-country Academy. In particular, they appreciated the opportunity to exchange best practices in the field of implementing new EU regulation such as MiCA and DORA. They also reported that real-world

[57] European Commission. Communication. "Digital Finance Package".

[58] European Commission. 9 March 2021. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee, and the Committee on the Regions. "2030 Digital Compass: the European way for the Digital Decade".

examples from financial supervisory authorities in other countries who have successfully managed risks and leveraged opportunities in the field of digital finance were very helpful. More broadly, the EU-SDFA has created a **community of financial supervisors** in the form of previous participants who have trained alongside each other and can now more easily consult each other as and when the need arises.

Best practices

First, multi-country flagship programmes are an effective way to provide technical support to many Member States needing support in areas that represent core EU priorities. From an organisational view, multi-country flagship programmes reduce the administrative burden for supervisory authorities as they facilitate the application process. They are particularly efficient in areas where the exchange of best practices is particularly important, as it is in the case of the EU-SDFA.

Second, with regards to the EU-SDFA, beneficiary authorities have suggested a few best practices. The longer, training modules, i.e. the **in-person courses** taught at EU level, both the foundation week and the advanced weeks were generally seen as most effective at strengthening Member States institutional and administrative capacity. This is partly due to fact that the longer in-person courses allow more time to delve into the course materials, and partly due to the fact that they allow more time to get to know fellow regulators and create a network. The **excellent quality** of the teaching itself and of the accompanying course materials, including PowerPoint slides, YouTube videos and the e-book broadened the reach of the EU-SDFA. Most **participants shared knowledge within their team**. The format of bringing professionals together in an in-person Academy can be seen as a success.

Suggestions for improvement

First, demand for training generally exceeded capacity. In the first two years of the EU-SDFA this was particularly true for the foundational courses and the available modules on MiCA and DORA. Beneficiary authorities would have appreciated being able to send more participants to these practical modules in particular.

Second, sharing information on the courses and, in particular, travel information earlier would make it easier for beneficiary authorities to decide whom to send and it would make it easier for selected participants to prepare. In this regard, it is important to note that the technical support providers have already acted on feedback from the first year. Since the beginning of the EU-SDFA's second year, the information sent to the national coordinating authorities and then to beneficiary authorities includes not just the course description but also the prerequisites, or the prior knowledge required to attend the course. Providing even more detailed information about the exact content of each course, perhaps already including a reading list, and the profile of the participants they are geared at, including, for instance, the job title or job description, would make it even easier to find the right staff to attend and to prepare them so as to maximise the individual, and organisational learning. Sending practical travel information early on will help participants plan their travels, and keep transportation costs down.

Third, beneficiary authorities saw differences in the relevance of the respective modules. While all topics were seen as helpful some were seen as more relevant than others. Some reported that the more practical modules were more relevant, or more urgent than the theoretical modules. As noted above, almost every beneficiary consulted reported that training on MiCA and DORA was particularly useful. Future iterations of the EU-SDFA may consider organising **more courses on new EU legislation that is relevant to supervisory authorities**.

One beneficiary authority interviewed for this study suggested that EU candidate countries would benefit from the EU-SDFA as well. Given that any new Member States will have to implement new legislation such as MiCA and DORA, and given, more generally, that the risks to the European financial sector extend beyond EU borders, this suggestion may be worth considering.

Similarly, given the success of the real-world examples shared in the EU-SDFA it may be worth expanding the peer-learning component of the teaching and encouraging even more national authorities to share best practices as part of the course.

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